

Sri Lanka Institute of Information Technology

Faculty of Computing

SE2012 - Object Oriented Analysis and Design

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Year 02 and Semester 01

Object Oriented Programming

- Object Oriented Programming is a method of implementation in which programs are organized as a collection of objects which cooperate to solve a problem.
- Allows to solve more complex problems easily.



Object Oriented Programming

- A complex system is developed using smaller sub systems.
- Sub systems are independent units containing their own data and functions.
- Can reuse these independent units to solve many different problems.

A Computer System



Object – General Meaning

object

noun

/ˈɒbdʒekt, ˈɒbdʒɪkt/ 

1. a material thing that can be seen and touched.

"he was dragging a large object"

synonyms: thing, article, item, piece, device, gadget, entity, body

Objects in the Real World



Grouping related things together



Classes

- We can classify objects into concepts. To do this we focus on the essential properties of an Object. Classes are Concepts.



Abstraction

- Distinguish between different Objects
- Classify Objects into Concepts
- Focus on the common properties



Abstraction

- Distinguish between different Objects
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Activity 1 – Identify Objects, Classes

- Lecturer
- Mr. Jeewaka
- Textbook
- Learning UML 2.0 from O'Reilly
- My copy of Learning UML 2.0
- A student
- A student at SLIIT
- A student in CSE 2nd year in 2025
- Person sitting next to you now!

Properties

- A class has a set of properties (attributes).
 - i.e. What do we need to store to describe a student?
- Activity - 2
 - i.e.
 - Name
 - Age
 - ...
 - ...

Activity - 3

- Payroll system

Class: Employee

What are the properties needed ?



Employee number Marital status Designation

OT Hours Age Loan Installment

OT Rate Basic Salary Height Name Allowance

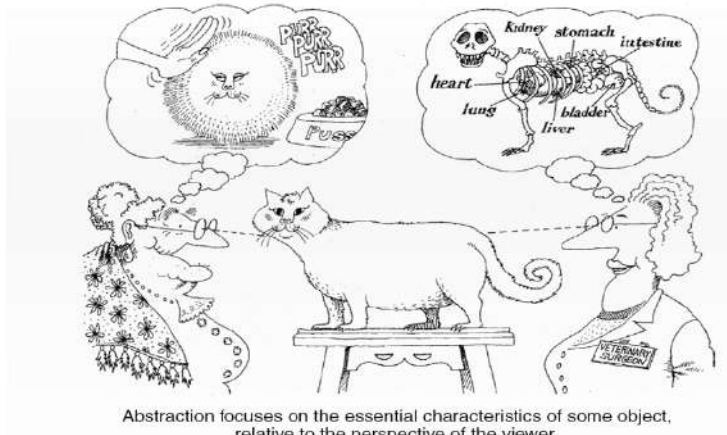
Address weight Bonus Number of children

Hobbies Insurance payment Favourite Movie

Abstraction

- Abstraction is the process of removing characteristics from 'something' in order to reduce it to a set of essential characteristics that is needed for the particular system.

Abstraction



Abstraction

- An abstraction denotes the essential characteristics of an object that distinguish it from all other kinds of objects and thus provide crisply defined conceptual boundaries, relative to perspective of the viewer.
- (Reference : Grady Booch, eta (2008), Object Oriented Analysis and Design with Applications 3rd Edition, pg 44)

Thank You!